2.2 Inverse Matrix

Say you solve $5x = 1$

$$\frac{1}{5} \cdot 5x = \frac{1}{5} \cdot 1$$

$$x = \frac{1}{5}$$

Typical enough but why does this work? Mostly because $\frac{1}{5}$ is the <u>multiplicative inverse</u> of 5, i.e. their product is 1.

Now say A and b are fixed matrices of appropriate size, and we want to solve $Ax = b$.

We did this earlier by forming augmented matrix $[A\ b]$

What if A had an "inverse" of sorts, denoted A^{-1} which "cancelled" A like $\frac{1}{5}$ cancelled 5. Then we could in theory instead left multiply both sides by this inverse to get:

$$A^{-1}Ax = A^{-1}b$$

---->

$$x = A^{-1}b$$

So could we maybe solve for x this way? Perhaps but much to clarify. For which A does such an inverse exist? And we know such systems sometimes have no solutions or infinite solutions. How would this work out in those settings?

Could be many motivations for this. Say bunch of linear systems $Ax = b$, $Ax = c$, $Ax = d$, and we want to solve all of them. Are we going to row reduce each augmented matrix?

We could but that seems rather annoying. If we had a way to "cancel" the A as alluded via inverse, might that be more efficient? Perhaps, again that's one of several motivations for this.

Consider $T: \mathbb{R}^n \dashrightarrow \mathbb{R}^n$ the linear transformation for which $T(x) = x$ for all x. This is the <u>identity mapping</u> where each point maps to itself, naturally only makes sense when domain and codomain are the same.

The associated matrix of this linear transformation is the <u>identity matrix</u>, denoted I_n is given by

$$I_n = \begin{bmatrix} T(e_1) & T(e_2) & \dots & T(e_n) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}$$

That is, the $n \times n$ matrix where every diagonal entry is 1, and every nondiagonal entry is 0 (this is one important diagonal matrix).

One big thing about this is for any $n \times n$ matrix A we have:

$$AI_n = I_n A = A$$

I_n behaves like the real number 1 in this context.

Important to write both orders to make this clear since as mentioned last time, matrix multiplication is not necessarily commutative.

There is naturally a different size identity matrix for each n, but nevertheless we often use I for identity matrix for simplicity under the expectation that the context makes it clear which size identity matrix we mean, should the size matter.

For inverse now consider linear $S: \mathbb{R}^n \dashrightarrow \mathbb{R}^n$. If there is an associated mapping $S^{-1}: \mathbb{R}^n \dashrightarrow \mathbb{R}^n$ for which

$$S^{-1} \circ S = S \circ S^{-1} = \text{identity}$$

that is called the <u>inverse</u> of S.

We really only try to define this for mappings from $\mathbb{R}^n$ to $\mathbb{R}^n$ not for $\mathbb{R}^n$ to $\mathbb{R}^m$ for $n \neq m$. Why?.

This was done on worksheet. If $n > m$, such a mapping cannot be one-to-one so can't possibly have inverse.

If $n < m$, such a mapping cannot be onto. So even though functionally it can have an inverse, that inverse won't have full domain $\mathbb{R}^m$ so this would not be the category of function we are studying right now, i.e. would not be a linear transformation whose domain is some full $\mathbb{R}^n$.

So we really only define inverse for square matrices. For A, $n \times n$, inverse denoted by A^{-1} satisfies

$$AA^{-1} = A^{-1}A = I_n$$

Say one of these two products is identity as needed. Is the other one guaranteed to be as well?

Might seem obvious, but that's not as clear as you'd probably think even though it is true.

Say instead B is $m \times n$ where $m \neq n$ and we have C $n \times m$ so that

$$BC = I_m$$

then will we necessarily also have

$$CB = I_n$$

Will not actually! Hence why for square matrices it's perhaps best not to just immediately assume it. This actually becomes quite important later in class, involves use of transposes.

Without fully proving that at this point, let's prove it in one direction. That is, for A some $n \times n$ matrix, suppose we "seem" to have inverse A^{-1} which satisfies

$$A^{-1} A = I_n$$

Just we are not sure if

$$AA^{-1} = I_n$$

is guaranteed. But let's say there is some B such that

$$AB = I_n$$

Hence why this won't be a full proof of what we are after since we are making the assumption that such a B exists which doesn't seem assured yet. But under that assumption, can we show that

$$B = A^{-1}$$

Start with

$$AB = I_n$$

and left multiply both sides by A^{-1}. When you do this kind of thing, be sure to specify on which side of the expression(s) you are multiplying (and in particular if it's an equation that you do it on same side with both!).

Again matrix multiplication is not commutative, that can make a big difference!

This gives

$$AA^{-1}B = A^{-1}I_n$$

----> $$I_n B = A^{-1} I_n$$

----> $$B = A^{-1}$$

So it does show that if we have a "left inverse" and a "right inverse" they are same. Just if there is say a left inverse, is right inverse guaranteed to exist? That's the unfinished business, can clear it up later when we have more tools.

Assuming inverse is safe that way, then it is unique. Why? Say A is a matrix with two inverses B and C. Then

$$I = AB = AC$$

----> $$AB - AC = 0$$

----> $$A(B - C) = 0$$

Since A has a "multiplicative inverse" we naturally know A is not the 0 matrix. So since this product is 0 and A is not 0 does that force B - C = 0? Not so fast!

Sure if these were real numbers that would be true, but two matrices can multiply to 0 without either being 0 (showed example last time). Key here is that A has an inverse. So left multiplying by that:

$$A^{-1}A(B - C) = A^{-1} \cdot 0$$

----> $$B - C = 0$$

----> $$B = C$$

This proves inverse is unique if it exists.

A square matrix which has an inverse is called <u>invertible</u> or <u>nonsingular</u> (naturally such a matrix which has no inverse is then called <u>singular</u>).

How to find those inverses? For 1 x 1, that's just real numbers. For 2 x 2:

Say

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The <u>determinant</u> of A is ad - bc. A is invertible exactly when $ad - bc \neq 0$. When it is we have

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Why? Well later tools can make it clearer where this came from. But for now, can verify this works via actually multiplying them, good enough for now.

Why does it make sense that the quantity $ad - bc$ being 0 versus not would be the key? Well if that quantity is 0 that means the rows of A are multiples of each other.

And once that is the case, the matrix will have at most one pivot so that there will be a nonpivot column guaranteeing that A, as a linear transformation, cannot be one-to-one so cannot have an inverse.

Ok, that clarifies why that quantity can be relevant for 2 x 2, how do we find something similarly useful in 3 x 3 and beyond? That's coming up shortly in the next chapter!

One use of inverse as alluded to earlier: say for A square and b proper size we want to solve

$$Ax = b$$

Especially think if we want to solve this for several different b's. Don't particularly want to row reduce many different augmented matrices as mentioned.

Ok the row reduction will be similar from one b to the next since A doesn't change. But still if A is invertible, left multiply by its inverse:

$$A^{-1}Ax = A^{-1}b$$

---->

$$x = A^{-1}b$$

Probably easier to compute $A^{-1}b$ for whichever b's you want it for rather than row reducing many times. This also indirectly tells us that whenever A is invertible this has a unique solution, no matter what b is.

Properties of Inverse

(1) $(A^{-1})^{-1} = A$

(2) $(AB)^{-1} = B^{-1}A^{-1}$

Why is (2) true? Note that

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

Thus AB is the inverse of $B^{-1}A^{-1}$ giving us (2)

So the inverse of product AB is actually quite easy to compute if A and B are both invertible and we know the inverse of each.

What about the inverse of A + B? Well that's not guaranteed to exist even if the inverse of A and B exist on their own (e.g. A = -B).

But even if it does exist, not necessarily related to A or B like the inverse of product is.

One of several reasons we use matrix multiplication more than addition, practical ways like that it's easier even if multiplying matrices directly is harder than adding them.

For 2 x 2 relatively simple formula for finding inverse, but what about 3 x 3 and beyond? Somewhat similar to dilemma with quadratic formula which is reasonably simple, helps solve quadratic equations.

But what about cubic equations and higher? Much harder, even impossible if degree too high.

With matrices, while as usual the calculation gets worse as the matrix gets larger, the actual strategy to find inverse does not change.

So finding inverse of an $n \times n$ matrix is actually a much easier question than generally solving an n-th degree polynomial equation. Like many things in linear algebra the size of the system does not affect the strategy, only the amount of calculation. Very relevant again if you are teaching a machine to do such things, as that obviously will have no trouble with the calculations.

One big key to finding inverses is elementary matrices.

An $n \times n$ matrix E is an <u>elementary matrix</u> if it can be obtained from I by doing one single row operation on I.

Important that it's only one for how these will be used! Recall replacing say $R1$ by $2R1 + R2$ is two operations not one! So doing that to I will NOT create an elementary matrix, and thinking it does could cause problems.

Many were doing this earlier when applying row operations to create row echelon form, that is doing two row operations at once without realizing it, thinking they were doing only one in that moment.

And even if they were doing that, though they might not have been thinking about it in exactly the right way, not like it was causing a problem since it still a valid calculation at end of the day.

But in the context of elementary matrices and in the next chapter, that can make a difference! Hence why I stressed that then as I didn't want you to think about it wrong from the get-go and then get in trouble when it really matters.

Big reason elementary matrices help is that if E is the elementary matrix created by performing one row operation on I, if we apply that same row operation to an arbitrary $n \times n$ matrix A, it will turn A into EA.

Fact that these row operations can be applied via using product is very useful for iterating the process.

Why is that so? Well can just test each of the three types of elementary row operations on an arbitrary matrix and compare that with the mentioned product to see that this is true if nothing else.

Every elementary matrix is invertible, in fact we can explain how to find its inverse. Say E is such a matrix.

Then a certain row operation will turn E into I (just reverse the one that turned I into E). If we apply that same row operation to I it will give us E^{-1}.

Why? Say E' is the matrix obtained by applying this row operation to I. Since it also turns E into I, by earlier idea we have $E'E = I$ which proves E' is E^{-1}.

With this in mind, note that any invertible matrix has its RREF the identity matrix for otherwise since the matrix is square, it would have a zero row in its RREF.

And once that is the case columns of matrix could not span $\mathbb{R}^n$ contradicting that every $Ax = b$ has a solution which must be true if A has an inverse as we computed the solution, $x = A^{-1}b$, earlier.

Thus invertible A has RREF as I so a sequence of row operations will turn A into I, let's say we have

$$E_k E_{k-1} \cdot \ldots \cdot E_2 E_1 A = I$$

via recalling that these operations can be applied by multiplying the matrix in question on the left by the appropriate elementary matrix. Rearranging gives us

$$A = E_1^{-1} E_2^{-1} \cdot \ldots \cdot E_{k-1}^{-1} E_k^{-1}$$

So then taking inverse of both sides gives us:

$$A^{-1} = \left(E_1^{-1} E_2^{-1} \cdot \ldots \cdot E_{k-1}^{-1} E_k^{-1}\right)^{-1}$$

And using inverse of product is product of inverses with order reversed (shown earlier) this simplifies to:

$$A^{-1} = E_k E_{k-1} \cdot \ldots \cdot E_2 E_1$$

Therefore applying the same row operations that turned A into I will turn I into A^{-1}.

Hence a way to compute A^{-1} is to take the "augmented" matrix $[A \mid I_n]$ which is n x 2n, turn the left half into I via row operations and then whatever the right half becomes is A^{-1}.

But honestly this can be fairly long. The smallest matrix you naturally might need this for is 3 x 3 which means working with a 3 x 6 augmented matrix. And considering you need RREF not just REF for the left half, can be long!

2.2.41 Find inverse of:

$$A = \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$$

So start with

$$\begin{bmatrix} A & I \end{bmatrix} = \begin{bmatrix} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 4 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{bmatrix}$$

R2 ----> R2 + 3R1 , R3 ----> R3 - 2R1 gives

$$\sim \begin{bmatrix} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & -3 & 8 & -2 & 0 & 1 \end{bmatrix}$$

Then R3 ----> R3 + 3R2 yields

$$\sim \begin{bmatrix} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{bmatrix}$$

Now REF so proceed to RREF via R1 ----> R1 + R3, R2 ----> R2 + R3

$$\sim \begin{bmatrix} 1 & 0 & 0 & 8 & 3 & 1 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{bmatrix}$$

Finally dividing last row by 2 so that the left half of the matrix is I yields our answer to be:

$$A^{-1} = \begin{bmatrix} 8 & 3 & 1 \\ 10 & 4 & 1 \\ \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{bmatrix}$$

2.3 Chacterizing invertible matrices

Many equivalent statements to an $n \times n$ matrix A being invertible, naturally depending on situation one thing might be easier to check than another. And just more general awareness of these equivalences helps you understand things better. One of those is:

(1) A has n pivot positions

Why equivalent? Well if A has n pivots, since it is square those pivot positions could only be the diagonal entries.

And thus the RREF would have to be I which we just showed means A is invertible, we in fact gave the algorithm for how to compute A^{-1} in that circumstance.

Other way, if A doesn't have n pivots, its RREF has at least one zero row which we said prevents A from being invertible since then some $Ax = b$ has no solution but each such equation must have a solution when A is invertible via left multiplying by A^{-1}.

Hence why A, $n \times n$, being invertible is equivalent to A having n pivot positions.

Next time will show more equivalencies to help us determine if a square matrix is invertible or not.